
Absence Is Not Omission: Availability, Vintage, and the Measurement of Selective Benchmark Disclosure

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Abstract

Providers choose which benchmark results a release document prints; independent evaluators score the same models on many more. Dating the instruments seems to answer whether omitted results trail printed ones: a benchmark published after a model shipped could not be reported. The comparison fails as a check on itself. Benchmarks postdating a release trail those predating it by 12.23 percentile points before any label enters, because the date deciding availability also fixes where a model sits in the benchmark’s comparison pool, an age-period-cohort collinearity reweighting cannot remove. The failure appears on a leaderboard we did not build: across fourteen HELM Lite releases, fixed models keep identical scores while every win rate moves; a pool-invariant companion moves nothing. Hand-coding every locatable release document, independently second-coded, shows the consequence: the coded gap rejects zero at +11.4, yet with labels removed the same releases return +14.1, the difference exactly what composition implies. A study built this way turns pool composition into apparent concealment. Reporting is performance-related, reported benchmarks standing 8.7 points above omitted eligible ones within release, yet no possession-grounded test detects withholding, the change-based gap keeping the predicted sign inside its null. Carrying a fixed suite across releases would make later silence informative; adding labels to a moving pool does not.

1 Introduction

For each frontier model release, providers disclose benchmark scores through a model card, while third parties execute standardised benchmarks for the same models [Liang et al., 2023, Epoch AI, 2026]. This paper asks whether benchmark availability can identify the older economic question of whether the provider’s undisclosed scores are typically worse than their reported scores. Only when the signal is common knowledge does disclosure unravel [Grossman, 1981, Milgrom, 1981]. As Dye [1985] details, when receivers cannot distinguish between senders who have no information, and senders who have bad information, receivers treat senders as the same, encouraging withholding. This indistinguishability extends to any research that attempts to analyse what was withheld, since we rely on the endowment, the signal a sender had at time of writing, which we do not observe in the work below.

The presence of benchmark release dates gives a rare ability to bound the endowment exogenously: when benchmarks are never shared before publication, any benchmark released after the model release is necessarily not in the provider’s endowment, regardless of whether providers ran it themselves. This lets us distinguish what was able to be disclosed, from what was unable to be disclosed.

First, we provide a new longitudinal study of benchmark disclosure that uses the boundary to construct a panel of a fixed snapshot of an independent benchmarking hub containing 819 model-versions, which we aggregate over scaffold variants to a panel of 353 provider releases, 46 organisations, 74 benchmarks, and 3,082 model-benchmark scores. We provide a verified release date for each benchmark, creating a new panel of 2,355 eligible model-benchmark scores, and 476 ineligible model-benchmark scores.

Second, we show that while the ability to bound the endowment is a unique and powerful feature to understand withholding, constructing a relative ranking of performance is required to judge withholding. Our central analysis compares reported, versus omitted benchmarks that were able to be reported, and requires the relative ranking to be unbiased across the two sides of availability. Instead, there is a 12.23 percentile point difference on average in performance between benchmarks created before the model release, versus those that were created after the model release, with pre-release benchmarks outperforming post-release benchmarks in 78.1% of the 146 releases that have cells in both groups. None of this calculation uses whether the provider reported the benchmark, and the least biased estimate of the relative ranking remains at 9.14 points. This remains a structural problem, as the verification of the endowment uses the same signal, the release date, that biases our relative ranking [Heckman, 1979]. In economic terms, the presence of benchmark vintages is a free signal to verify the endowment, which Dye [1985] demonstrates cannot be known to receivers in general. However, the signal to verify the endowment is not excludable in the relative ranking, and therefore is self-contaminating. This raises a fundamental design challenge for any research that verifies the presence of information via vintage, but then prices evidence using a relative ranking, win rate, or percentile. Not only must reporting decisions be observable, but also the relative ranking must maintain a consistent reference group that is invariant to the endowment.

In our third finding, we extend these concerns beyond our panel. In a sample of 14 published releases for HELM Lite, the pool expands (30 $\rightarrow$ 91 models); 24 models keep bit-for-bit identical scores; and all their published headline values change (five pairs reversing order). These changes do not show up in a parallel leaderboard maintained by the same group of authors, which presents an absolute headline. of HELM Lite, the pool grows from 30 models to 91 while 24 models keep bit-identical scores, and every one of their published headline values moves, five pairs reversing order. A companion leaderboard from the same team, behind an absolute headline, shows none of this.

In our fourth finding, we code this history. We hand-code the 100 releases with a locatable release-time document, independently replicated on a fifth ($\kappa = 0.95$ on reported vs. not), yielding the estimator such a study would use. Omitted benchmarks are +11.4 points higher than those the provider could not have reported (excluding zero, but falling short of its matched counterpart by what composition implies: it cannot separate concealment from the availability gap). What survives is composition: reported benchmarks are 8.7 points higher than omitted eligible benchmarks within release ($p = 0.0001$), and this is robust to benchmark fixed effects.

1.1 Related literature

The state of reporting in model documentation is imperfect, incomplete, and dependent on the sender, established [Stauffer et al., 2025, Kuehnert et al., 2026], with two papers focusing on the reporting gaps themselves: Zhang et al. [2026] finds a benchmark family missing from all main result tables across four flagship releases, and Ghosh et al. [2026] defines the reporting standard that would make the omission observable. Both lack benchmark dates, so there’s no way to distinguish a skipped benchmark from one the release predates. Bean et al. [2025] instead evaluates 445 benchmarks for construct validity, a different question about whether instruments measure what they’re claimed to, while we take the instruments as given.

Evidence is stronger on leaderboards: Singh et al. [2025] characterises selection in real time (providers testing private variants and publishing the best), while Xu et al. [2025] shifted HELM’s primary metric off a win rate that depends on the models compared. In both cases, the leaderboard changes for reasons other than the evidence, but neither isolates, and we do at the cell level, an omission from a result that could not have existed. What no paper

here controls for is availability, which we introduce: absent a confirmed benchmark date, the distinction between “wasn’t reported” and “did not exist” collapses.

The economics literature frames the problem and gives a cautionary note: silence unravels when endowments are commonly observed [Grossman, 1981, Milgrom, 1981] but persists if the recipient cannot distinguish an empty endowment from a concealed one [Dye, 1985, Jung and Kwon, 1988] or when revelation incurs a cost [Verrecchia, 1983], frictions our design does not separate. Empirically, the warning bears out: unravelling doesn’t occur in practice [Dranove and Jin, 2010, Jin and Leslie, 2003, Hotz and Xiao, 2013, Bederson et al., 2018] and in the lab [Jin et al., 2021]. What all this work misses is the endowment directly, the thing our bound aims to recover. Meanwhile, concurrent theory takes a direct approach to strategic assessment and persuasion [Truong et al., 2026, Dai et al., 2026]; theory, not measurement.

2 Data and Setting

2.1 The disclosure problem

A release i ships at date t_i carrying a single document. Let $\mathcal{B}$ index benchmarks, each with its own publication date τ_b , let $E_i \subseteq \mathcal{B}$ be the endowment, the benchmarks whose result the provider holds when it writes, and let $D_i \subseteq E_i$ be the disclosed set, which reports a score s_{ib} for each $b \in D_i$. The receiver observes D_i and its scores and must split the complement into results held back and results never held.

Under a commonly known endowment, silence is priced at the worst value it could conceal and disclosure unravels to $D_i = E_i$ [Grossman, 1981, Milgrom, 1981]. Two distinct frictions break that. In Verrecchia [1983] the endowment is known and disclosure is costly, so withholding is a cutoff in s_{ib} . In Section 3 of Dye [1985] disclosure is free but the receiver cannot tell an empty endowment from a concealed one, because a sender can prove possession and cannot prove emptiness.

Here an independent evaluator publishes s_{ib} for cells the provider passed over, and each τ_b makes infeasibility verifiable. Writing $A_i = \{b : \tau_b < t_i\}$ for the available set and O_i for the cells the evaluator scores,

$$D_i \subseteq E_i \subseteq A_i, \quad \theta = \Pr(b \notin D_i \mid b \in O_i \cap A_i).$$

A provider may hold results the evaluator never produced and a benchmark can exist and go unrun, so θ is defined against the evaluator’s coverage rather than the true endowment, and the Dye friction survives inside A_i .

Judging what was withheld then requires a measure of standing P_{ib} , since raw scores are not comparable across benchmarks. The design rests on an exclusion restriction: conditional on the innocent explanations, membership in A_i does not enter the equation for P_{ib} [Heckman, 1979]. A date rule cannot satisfy it: $t_i - \tau_b$, t_i and τ_b are exactly collinear, so if any two of the three move P_{ib} the third cannot be excluded from it. All three move it here, which Section 3 shows and Appendix B bounds. Two things follow, and they are not the same. The first is practical: estimating θ needs disclosure labels, which the coded record of Section 5 supplies. The second is structural: reading θ as evidence of withholding runs into A_i itself. Availability is a deterministic threshold in benchmark maturity, $\mathbf{1}[t_i - \tau_b > 0]$, so no cell of the same maturity sits on both sides of the boundary. Wherever standing depends on maturity, conditioning finely enough to make availability excludable destroys positivity. So availability alone cannot globally identify selective disclosure from any pool-relative standing measure, win rate or percentile alike, and a jointly estimated scale inherits the same selection through which cells exist. What survives, under continuity, is the discontinuity at the threshold.

2.2 The independent score set

We work with a pinned version of Epoch’s benchmarking hub scraped on August 17th, 2026. This data includes 819 model-versions. Since we are interested in release-level disclosure, we use organisation, model and release date to identify releases. Epoch uses a version attribute to distinguish different scaffolds within the same release, e.g. different reasoning effort or context length variants. Since the provider publishes one document for one release, we

collapse these versions to releases, taking the maximum score among scaffolds and retaining the spread as a check. We end up with a dataset with 353 releases, 74 benchmarks and 3,082 release-benchmark cells that have an associated independent score. The data is clustered by provider.

2.3 Eligibility and benchmark dates

A provider cannot omit a benchmark that did not yet exist. To ensure temporal eligibility, we collect release dates for benchmarks: the earliest verifiable public release date (from Epoch if available, from the primary source of the benchmark if not, translating between preprint dates, code repository dates and different versions to get the first public release date). A cell is eligible only if the benchmark release date strictly predates the release’s date and if the benchmark’s release date is verified. This restriction assumes providers could not have access to unreleased benchmarks; if a provider was given early access to a benchmark, this assumption would not hold. Such violations shrink rather than create the reporting gap; they move available cells into the postdating category. Such violations leave no trace in our coded data: no document reports a benchmark before its public release date. This restriction leaves 2,355 out of the 3,082 cells eligible. Cells whose benchmark strictly postdates the release’s date form a separate set of 476, cells that could not have been reported. We drop cells without a verified benchmark date rather than imputing a date; inaccuracies in dates shrink the boundary estimate we construct in Section 3. For comparing releases, we do not require a minimum number of benchmarks; this would exclude the releases that only have one eligible and one postdating cell. 57 out of 146 releases with both eligible and postdating cells have fewer than eight scored benchmarks. The more trivial obstacle, that how many benchmarks a release could report is set by when it shipped, is discussed in Appendix A.

To code disclosure, we adapt ORBIT, Outcome Reporting Bias In Trials. ORBIT is standardised in clinical trials, where disappearing outcomes (outcomes specified in the protocol but not reported in the publication) have a long literature on measurement [Chan et al., 2004, Kirkham et al., 2010, Dwan et al., 2013]. For existing cells, we record whether the provider reports a number, cites the benchmark without reporting, or reports a variant; for missing cells, we record whether a previous member of the same family reported it, if contemporaries reported it, or if no similar provider reported it. Of the 108 releases in the queue, 103 with eligible cells in our pinned dataset and five in a more recent vintage, we code the 100 for which we can locate a document from release time; 1,280 cells in total. None of the analysis before Section 5 depends on coded disclosure.

2.4 Measuring standing

Raw scores do not allow comparison across benchmarks, so we report relative performance as a percentile within benchmark by comparing each model to models whose release dates fall within 182 days before or after the focal release (“window”), using midranks for ties. We also report the number of peers and their share of the window released later than the focal model. This is the metric implied by a windowed comparison and what we demonstrate to be biased in a way that mimics selective disclosure.

2.5 Inference

Inference is constrained by the same structure. The panel has 46 nominal provider clusters, but 24 contribute fewer than ten cells, 20 appear in a single release, and the five largest hold 68.1% of the sample. Cluster-robust asymptotics are not credible below twelve clusters, and we report no analytic standard error there [Cameron et al., 2008]. Regression coefficients in that range use the restricted wild cluster bootstrap over 9,999 draws, which imposes the null before resampling Rademacher weights at the provider level. Release-level means use a cluster sign-flip randomization test over 9,999 draws, reversing an organisation’s contribution at once; it preserves provider dependence but not benchmark dependence, which the two-way clustered regressions carry, and their intervals are percentile bootstraps over 10,000 provider resamples. Both report $(k + 1)/(\text{draws} + 1)$, with k the draws at least as extreme as the observed statistic. Mirrored sign vectors leave 2^{G-1} distinct statistics over G clusters, so a resolution of 0.0001 needs fifteen clusters; the 18 organisations behind the headline estimate

clear it. Below twelve clusters the bootstrap switches to a six-point weight distribution whose finer support restores it [MacKinnon and Webb, 2016, 2018].

3 The Availability Gap

3.1 The placebo comparison

To separate withheld from non-existent, we require a control group of non-endowment [Dye, 1985]. Dates appear to provide one: a postdating benchmark could not be reported without prepublication access, whereas an eligible one could. A disclosure label would report whether it was, and none enters. Any innocent reason an eligible looks poor, difficulty drift, saturation, or measure change, operates equally well on both groups, meaning their difference should be zero. One such innocent reason is inherently asymmetric: a benchmark prior to shipping can lie in the training set for a given model and one after it cannot, so contamination increases eligible standing only, beyond any reweighting of the comparison. Not zero. On average eligible beats postdating by 12.23 percentile points among 146 releases of both cells, with the median 10.69, and the proportion of releases where it wins is 78.1%. This computation does not know whether benchmarks are reported. By constructing one that treats both sides of the window equally (see below), we find eligible to beat by 9.14 percentile points across 141 releases, positive 70.9%, still without removing it. Using a cluster sign-flip randomization test of the difference at the 18 organisations with both cell types produces $p = 0.0001$ as the smallest available from the draws. Groups ineligible to be reported already stand lower than ones eligible, so it is not a control.

3.2 Decomposing the gap

In Table 1 we identify the failure using a placebo regression with an indicator for whether the benchmark postdates the release (always within release). With release effects alone this is -13.5 points. But benchmark fixed effects account for 63.4% of it, structurally since the placebo cells are constructed to exist primarily in those benchmarks arriving late, which are less saturated, with 8.3% of scores in the latter half of the 54 benchmarks with ≥ 10 scores in the top decile of its own range vs. 13.2% in the former ($p = 0.015$), leaving more room below the ceiling. The term capturing the asymmetry in peer-window, i.e. what share of comparisons are made to models released after the model of interest, controls for 39.8% on its own, and once we include both benchmark fixed effects and both window terms, the remaining is 0.003 points, with a confidence interval up to $+3.4$, which is no more than one third of the pre-control difference.

Conditioning set	Placebo coefficient	t	Absorbed
Release fixed effects only	-13.529 (2.038)	-6.637	0.0%
Release FE + benchmark fixed effects	-4.949 (2.183)	-2.267	63.4%
Release FE + peer-window asymmetry	-8.151 (1.865)	-4.370	39.8%
Release FE + peer count	-2.107 (1.882)	-1.120	84.4%
Two-way fixed effects + both window terms	-0.003 (1.748)	-0.002	100.0%

Table 1: Placebo coefficient by conditioning set, percentile points, within release. Two-way provider-by-benchmark standard errors (45 by 61 clusters; provider-only in Appendix B), $n = 2,831$. Absorbed is the fall in absolute value from row one; peer count alone suppresses; the last row has both.

For the second dimension, the mechanism is geometric, with the peer-window being symmetric in calendar time, 182 days before or after, but not model coverage since the coverage begins when the benchmark arrives. Figure 1 depicts both objects against benchmark maturity at release: the relative proportion of newer peers declines with maturity and jumps at the availability threshold, whereas standing runs through it without a detectable break. Thus, the mechanism is discontinuous where the outcome is not; composition is a property of timing alone.

The share of a cell’s window released after the focal model is 0.4683 for eligible cells and 0.7065 for placebo cells, against 0.5 for a two-sided window. Within release, the slope of standing on that share is -30.63 percentile points per unit among eligible cells alone. We plot standing versus the share of newer peers simultaneously for both types of cells

in Figure 2. Instead of the two types of cells forming distinct point clouds, they follow a common downward trend that only appears after accounting for changes in the benchmark makeup (see Table 1); eligible and placebo cells lie at different points on that line. By construction, eligible and placebo cells are separated and never overlap on their assignment variable, the maturity of the benchmark at release; thus the difference between eligible and placebo cells is obtained by extrapolation rather than comparison on the common support. As a robustness check, we probe the continuity of the regression at the cutoff, the only part of the assignment variable where the analysis is identified, see Section 2.1 [Hahn et al., 2001]. The regression discontinuity analysis does not reveal a jump at the cutoff. Local linear regressions give negative coefficients for all bandwidths and orders, all confidence intervals include zero with the upper bound of the widest interval being +8.05, and the difference on a balance-driven bandwidth is -2.52 with a randomization p -value of 0.447 [Cattaneo et al., 2015]. The two types of cells differ in mean maturity by 793 days, and the discrepancy between them is driven by that distance rather than the cutoff point itself (Appendix B for the balance tests).

It’s possible to decompose standing as a weighted combination of percentile-ranks against older and newer peers with the share of newer peers as weight, percentile = $(1 - w)P_{\text{old}} + wP_{\text{new}}$ (exact cell-wise). The side-balanced standing, setting the weight to one half, available for 92.5% of all cells (92.8% for eligible-or-placebo cells), cuts the same slope to -8.56 .

Despite the identity holding, the gradient is not a mere book-keeping tautology: permute the scores within the same benchmark, and the capability trend disappears while publishing dates, window membership, peer counts and newer shares are bit-wise identical. Under that null the slope has an average of -0.09 with a standard deviation of 3.12; the measured value of -30.63 is 9.8 standard deviations out, unmatched by any of the 300 draws. Further, fitting a linear regression within each benchmark of scores against publishing date, then reconstructing the scores as predicted values plus permuted residuals within each benchmark, meaning there’s only a capability trend but no others, we are able to recover 93% of the slope over 200 draws. The comparison-set geometry by itself is inert; it only introduces bias when combined with the existing trend in capabilities.

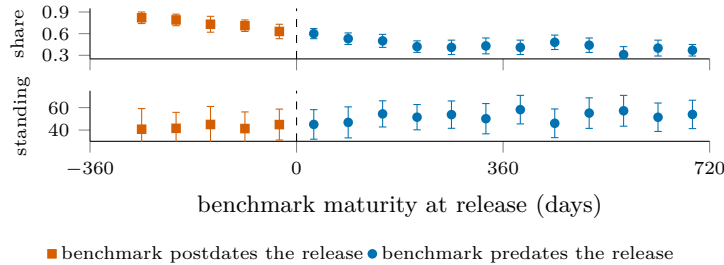


Figure 1: The mechanism and the outcome at the availability boundary. Top: the share of a cell’s window released after the focal model, against benchmark maturity at release; it falls with maturity and jumps at the boundary. Bottom: within-benchmark standing on the same axis, continuous through it. Sixty-day bins; intervals clustered on organisation.

3.3 Trimming and reweighting

The phenomenon has a name in nonparametric estimation: local averages are biased near the edge of a support, and the standard remedies recentre rather than discard [Gasser and Müller, 1979, Fan, 1992]. What is unusual is the boundary’s coincidence with the endowment: a benchmark was available to a release exactly when the model did not sit at the left edge of that benchmark’s coverage. The indicator of what a provider could have held is therefore the indicator of where the window is one-sided, and removing the edge removes the contrast. The coincidence has an older name, the age-period-cohort identity: release date, benchmark vintage and their difference are exactly collinear, and each has a measured signature here: t_i in the capability trend, τ_b in saturation, $t_i - \tau_b$ in window composition. Only the linear components are unidentified in that structure [Fosse and Winship, 2019a,b], so what fails is a slope. Three sign restrictions bound it: capability rises with the calendar, later benchmarks are less saturated, and standing rises with maturity. All three are assumptions on unidentified components, each motivated by a measurement of

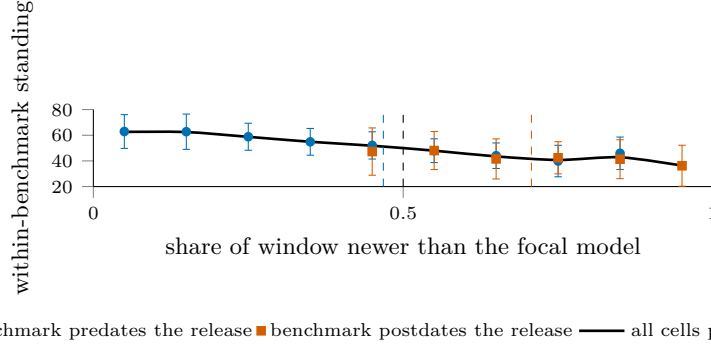


Figure 2: Standing against the share of a cell’s window released after the focal model; dashed verticals mark group mean shares, the black dashed line the symmetric 0.5. Eligible and placebo cells lie on one declining curve, so the contrast between them is a position on the curve, not a fact about disclosure. Intervals clustered on organisation.

an identified quantity. Under them the unidentified component accounts for at most 24 to 34% of the placebo gap across specifications, roughly three percentile points, which is where the saturated regression also lands. The remainder is carried by the identified reduced-form slopes, and Appendix B gives the derivation.

Standing measure	Mean	Median	Positive	Sign-flip p	Releases	Cells
Windowed percentile, $\pm 182d$	+12.23	+10.69	78.1%	0.0001	146	100.0%
Rank against all models ever	+22.75	+22.66	89.0%	0.0001	146	100.0%
Trim to two-sided windows	+9.96	+10.69	72.2%	0.0001	115	70.4%
Side-balanced percentile	+9.14	+9.62	70.9%	0.0001	141	92.8%

Table 2: The label-free placebo null under each candidate measure: release-level mean standing of eligible minus postdating benchmarks, in percentile points. Every entry should be zero. Sign-flip p is the provider-clustered test of Section 2.5; Cells, the share of cells carrying the measure.

Table 2 shows what follows. Ranking against every model ever scored nearly doubles the contamination; trimming to two-sided windows discards 30% of the cells to remove 19%, its row the uncorrected gap on the retained cells, composition rather than correction. Shortening the window does reduce the gap, to +8.59 at 91 days, but in proportion to the asymmetry it removes rather than by breaking the mechanism. Across four widths and four benchmark-count minima the gap is positive in all 16 specifications, from +8.59 to +21.66, and tracks the eligible-minus-placebo gap in window composition at $r = 0.991$. Dropping any one of the 18 organisations leaves it between +11.45 and +13.20. The correction is a covariate, not a subsample.

One reading of that number would be too generous to it. Window composition is a function of the same dates that define the placebo indicator, so conditioning on it does not adjust for a confounder sitting outside the comparison; it splits the gap into the channel carrying it. A study that holds disclosure labels and wants an estimate of θ must first argue that window composition is settled before a provider decides what to report, or it conditions away part of what it measures. Provider identity explains little of the variance in window composition; Appendix B reports the decomposition and the coverage tilt it does not correct.

The side-balanced percentile is the one change of measure doing substantial work, cutting the placebo mean to +9.14 and the eligible-cell gradient to -8.56 . Figure 4 in Appendix B puts it against the uncorrected measure, near flat where cells are dense. Reweighting cannot reach what remains: balancing repairs the weight on each side of a model but not which models an evaluator chose to score early, so evaluator attention stays a confound. Undefined wherever a side is empty, it accompanies rather than replaces the uncorrected measure.

4 Evidence from HELM

Everything so far concerns a percentile constructed here on a panel assembled here, so the effect may belong to the construction, not leaderboards in general. HELM Lite publishes its full accuracy table at every versioned release, archives earlier ones, and reports a pool-

relative headline [Liang et al., 2023], so the same evidence reads at 14 points in time while the pool grows.

If a leaderboard moves, the first thing to rule out is a revised suite. Across releases v1.0.0 through v1.13.0, the ten scenario columns are identical, with nothing added, dropped or renamed beyond one relabelled column header. The same 24 models appear in all 14 releases with their 240 published scenario scores bit-identical throughout, while the evaluated pool grows from 30 models to 91.

Against that frozen evidence, all 24 constant models see their published Mean win rate change, by 0.1232 on average in absolute value, up to 0.2178 for Mixtral 8x7B. Most of that movement is arithmetic: Mean win rate is the fraction of head-to-head comparisons a model wins, so a fixed model faces a larger, later field as the pool grows; the change correlates with initial standing at -0.766 .

The movement that matters is elsewhere. If the drift were a monotone transform of the frozen evidence, it would be harmless, since a monotone rescaling preserves every order. It is not. Of the 24, 22 fall and two rise, and five of the 276 pairs among them, 1.8%, reverse between the first release and the last, each a published order inversion with bit-identical evidence beneath it. Figure 3, in Appendix B, plots each model’s rank among the constant set at the first release against the last, so a reversed pair is a crossing, countable on the page while the evidence beneath is fixed. Unlike test-set reuse [Roelofs et al., 2019], the evidence is fixed; the reference set is what moves. Singh et al. [2025] reports a comparable pattern from a different mechanism, deprecation policy rather than a coverage boundary changing which models remain available, and in both the reference set rather than the evidence moves the ranking.

The control, HELM Capabilities, comes from the same organisation, whose maintainers had diagnosed the win rate as pool-dependent [Xu et al., 2025]; it runs on the same infrastructure and cadence, its pool growing comparably, from 22 models to 68 over 16 releases. Its headline is an absolute mean over scenarios rather than a fraction of comparisons won. The same 22 models appear in every release with 110 scenario scores bit-identical, none of the 22 headline values moves, and no pair reorders, at the endpoints or ever. An absolute mean over bit-identical scores on a fixed suite cannot move with the pool, so zero drift is arithmetic, a scope condition rather than a test that could have failed.

This need not be read as an error on HELM’s part: the absolute headline on its own Capabilities leaderboard is the point. A pool-relative headline is a joint statement about a model and about the population evaluated beside it; read later as statements about the models alone, two such numbers move standing between providers for reasons belonging to neither. Restoring comparability across moments is the classical equating problem [Kolen and Brennan, 2014], avoided by a scale estimated jointly [Ho et al., 2025]. Panel and leaderboard fail identically: the reference set, not the evidence, moves the number.

5 Coding the Record

A benchmark can be missing from a model card because it was irrelevant, never run, or run and withheld. Only the third is disclosure in the sense of Grossman [1981], Milgrom [1981] and Dye [1985], and only it is a reader’s to price. The within-family drop narrows it: a provider that reports b at t and omits it at $t + 1$, an independent score present, has established relevance and possession of the harness, the independent score its registry-style counterfactual [Chan et al., 2004, Dwan et al., 2013, Kirkham et al., 2010]. It does not establish that b was run for $t + 1$, so the Dye [1985] uncertainty narrows rather than closes.

A single coder read the release-time document of each of the 100 releases for which one is locatable; eight of the queued 108 have none. The record notes every benchmark a document reports, with variants, bare mentions, and stated benign absences; the ORBIT categories are derived mechanically from these records and family history. This yields 1,280 coded cells; 31 from five later-vintage releases sit outside the pinned panel and are excluded. A second coder then recoded a provider-stratified fifth of the releases, blind to the first coding and without calibration contact; Appendix B records the one release where that blindness broke. Agreement is Cohen’s κ over the derived cells [Cohen, 1960], with intervals from resampling releases. The two codings agree at 0.95 [0.80, 1.00] on the primitive judgment

of whether a benchmark was reported, at 0.85 [0.72, 0.95] over the full category set, and at 0.81 [0.48, 1.00] on the high-versus-low-suspicion split. The single E cell each coding derives is the same cell, encouraging but no reliability estimate for so rare a category; Appendix B reports the confusion structure. Among the 1,249 eligible coded cells that resolve in the panel, 76.9% report no number, and the release-level mean is 73.1% [65.9, 79.2]. The rate is descriptive, not by itself evidence of selection.

Standing measure	Coded gap	95% CI	No labels	Difference
Windowed percentile, $\pm 182d$	+11.40	[+5.6, +17.0]	+14.13	-2.73
Side-balanced percentile, $\pm 182d$	+7.90	[-0.4, +15.2]	+9.82	-1.92
Rank against all models ever	+22.33	[+18.0, +28.1]	+24.50	-2.17
Windowed percentile, $\pm 90d$	+8.15	[+1.3, +14.7]	+10.36	-2.21
Windowed percentile, $\pm 365d$	+18.03	[+13.9, +23.1]	+20.53	-2.50

Table 3: The coded omission deficit beside the label-free gap, percentile points, over 71 releases and 13 providers (70 side-balanced). No labels is the contrast with no disclosure coding; Difference is Coded gap minus No labels. Provider-clustered bootstrap intervals; four of five exclude zero.

Table 3 is the section’s result. The disclosed-versus-omitted comparison returns +8.7 points [4.1, 13.9] over 63 releases; the intended availability contrast, omitted-eligible against post-dating cells over the 71 releases holding both, returns +11.4 [5.6, 17.0] and would reject a zero null at $p = 0.003$. Read against Section 3, it is the availability gap: the label-free contrast on the same releases is +14.1, and under every measure and window the coded gap falls 1.9 to 2.7 points short. The shortfall is an exact accounting identity, the average over the same releases of the reported share times the within-release reported-omitted gap, +2.7, so the labels change this contrast only through composition.

What survives is composition. Reported benchmarks stand 8.7 points above omitted eligible ones within release, a gap larger than all 9,999 count-preserving random assignments ($p = 0.0001$, the draws’ floor; null standard deviation 1.9) and one the availability gap cannot produce. It survives coding-error, variant-reclassification, and leave-one-out exercises because the availability gap does; benchmark identity does not carry it, since benchmark fixed effects beside the release effects raise the coefficient from +7.4 to +9.3 and the window-share control leaves +7.4. It does not identify whether standing drove the reporting: relevance, salience, and within-eligible contamination can move disclosure and performance together.

Possession is the remaining distance; the original drop estimator and four possession cuts return one verdict. The drop estimator realises 16 drops across 15 releases and returns +2.9, opposite the withholding prediction, sign-flip $p = 0.25$ over five providers, the null its calibration priced. No document states that a benchmark was evaluated while reporting neither a score nor an outcome, across the 295 cells documenting a run. A change-based test, each drop against its standing under the previous release, gives -3.7, the predicted sign, permutation $p = 0.16$ over 42 transitions; possession tiers and equal-sized substitutions detect nothing further. Appendix B details these and three unrepaired limits: unlocatable documents, coverage that may follow provider emphasis, and best-across-scaffolds scoring.

6 Conclusion

We set out to measure how much of what could be reported goes unreported by choice. The coded record returns the number and the reason to distrust it: +11.4 points, rejecting zero yet short of its label-free counterpart by what composition implies, too imprecise to exclude a strategic component. Such a study turns composition into apparent concealment. Three claims separate: non-reporting is extensive, composition is performance-related, withholding not established, possession tests returning null, absent, or opposite-signed evidence. Dating instruments bounds the endowment, an access assumption, fixing where a model sits in a benchmark’s coverage: what identifies it moves the measure pricing silence. A joint scale [Ho et al., 2025] shrinks but does not close the placebo gap; a fixed suite carried across releases would make later silence informative, narrowing the Dye [1985] pooling and growing the 16-drop record.

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520 A What a Provider Could Report

521 The set of results a provider could publish is not fixed, and it grows. When we count the
 522 benchmarks in our panel by the year each was published, the dated stock rises from 11 at
 523 the end of 2019 to 62 by 2026. A benchmark that enters that stock stays available to every
 524 release after it, so the denominator of any omission rate is weakly increasing by construction.

525 What a single release could report grows more slowly and less evenly. Averaged by half-year,
 526 the number of eligible benchmarks per release runs 6.8 in the first half of 2023 and falls to
 527 3.9 in the first half of 2024, a period in which releases outpaced the instruments now used
 528 to score them. By the first half of 2026 it reaches 10.1. The two series move together over
 529 the sample and apart within it, which is consistent with the opportunity set being shaped
 530 partly by release timing rather than by provider choice.

531 A count of omissions therefore measures the calendar. Unless reported tables expand one
 532 for one with the stock, omission rates drift upward for arithmetic reasons alone, and com-
 533 paring early releases with late ones on that measure compares eras rather than providers.
 534 The question that survives is composition: which benchmarks are dropped, conditional on
 535 availability, against what an indifferent selection over the same set would have produced.
 536 Answering it requires knowing not only what was available but how the release stood on
 537 each available benchmark, and it is that second requirement, not the first, that the rest of
 538 the paper shows to be the hard one [Dranove and Jin, 2010].

539 B Additional Checks

540 Is window composition a provider’s choice? Section 3.3 conditions on window composition
 541 and notes that doing so is defensible only if composition is settled before a provider decides
 542 what to report. Regressing the share of a cell’s window released after the focal model on
 543 provider dummies gives an R^2 of 0.071 across 45 organisations, against 0.002 for benchmark
 544 dummies over 61 benchmarks and 0.104 for the release date alone. Window composition is
 545 close to idiosyncratic at the cell level, which is consistent with its being set by the evaluator’s
 546 scoring schedule rather than by the provider, and is not enough to establish it.

547 Is the evaluator’s coverage selective? The estimand conditions on the cells the evaluator
 548 scores, so coverage is upstream of everything else. Coverage averages 8.7 benchmarks per
 549 release with a standard deviation of 8.0 and a range of 1 to 38. It correlates with a provider’s
 550 release count at 0.203 and with the release date at 0.265, and releases from the densest
 551 quartile of providers carry 9.6 scored benchmarks against 8.4 for the rest. Coverage is
 552 therefore mildly tilted toward larger and later providers, and nothing in this paper corrects
 553 for that.

554 Bounding the unidentified component. Write t for a release date, τ for a benchmark’s
 555 vintage and $a = t - \tau$ for the benchmark’s maturity when the model met it. A regression
 556 of standing on all three is singular by construction, which is the age-period-cohort problem,
 557 and the data identify only $\Pi_p = \beta_a + \beta_t$ and $\Pi_c = \beta_\tau - \beta_a$, leaving a one-parameter family
 558 indexed by $\lambda = \beta_a$. Fitting the reduced form on the 2,831 comparable cells gives $\Pi_p = +4.12$
 559 and $\Pi_c = -1.36$ percentile points per year.

560 Three restrictions come from the paper rather than from outside it. Capability rises over
 561 calendar time, so $\beta_t \geq 0$. Later benchmarks are less saturated, which Section 3.2 measures
 562 directly, so $\beta_\tau \leq 0$. Standing rises with maturity, since a model on a young benchmark is
 563 ranked against newer peers, so $\beta_a \geq 0$. Together these give $0 \leq \lambda \leq \min(\Pi_p, -\Pi_c) = 1.36$,
 564 and saturation rather than capability is the binding restriction. The mean maturity gap
 565 between eligible and placebo cells is 2.17 years, so the unidentified component can account
 566 for at most $1.36 \times 2.17 = 2.94$ percentile points, 24% of the observed gap. Adding centred
 567 quadratic and cubic terms in t and τ , which are not collinear with a , moves the ceiling to
 568 34% and 31%.

569 Shuffling scores within benchmark leaves every date and the rank deficiency untouched while
 570 destroying the capability trend. Under that null the same quantity falls to 0.02 percentile
 571 points on average over 30 draws, with a maximum of 0.24, so the bound reflects the trend
 572 the geometry multiplies rather than the geometry alone.

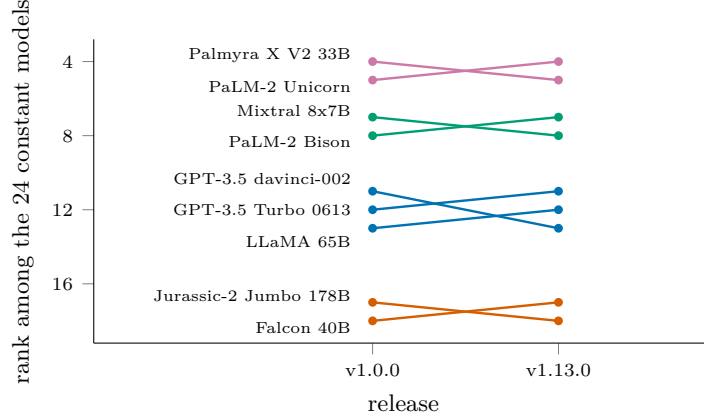


Figure 3: The nine HELM Lite models whose published order changed between the first release and the last, ranked among the 24 whose 240 scenario scores are bit-identical throughout; coloured by group of mutually reversing models, four groups and five reversed pairs in all. The control: across 16 HELM Capabilities releases behind an absolute headline, with the pool growing from 22 to 68, none of the 22 constant models’ headline values moved and no pair reversed.

573 The boundary as a discontinuity. Eligibility is $1[t_i > \tau_b]$, a deterministic function of the
574 two dates, and that is what the global repairs run into. Equating across groups that took
575 different instruments assumes a conditional relationship common to both, which two groups
576 defined by the boundary cannot have [Sinharay and Holland, 2010], and reweighting by
577 the probability a cell is observed needs that probability bounded away from zero, which a
578 deterministic rule denies [Ma and Chen, 2019]. The same determinism makes the boundary a
579 sharp discontinuity, where standing at the threshold is identified under continuity whatever
580 happens globally [Hahn et al., 2001]. Local linear estimates with a triangular kernel and
581 provider-clustered errors give -8.82 (4.76) at ± 60 days, -6.85 (4.09) at ± 90 , -4.08 (3.11)
582 at ± 180 and -2.85 (2.35) at ± 250 , and a quadratic fit at each width moves none of them
583 across zero. All eight are negative, all eight intervals cover zero, and the largest upper bound
584 is $+8.05$, below the $+12.23$ the global comparison reports.

585 The local randomization framework agrees [Cattaneo et al., 2015]. Widening symmetric
586 windows and testing the release’s benchmark count, scaffold count and scaffold spread for
587 balance, we find the smallest of the three p -values to be 0.335 at ± 20 days and 0.571
588 at ± 40 before falling to 0.058 at ± 60 , which selects ± 40 , a rule applied after seeing the
589 balance results, so the randomization p below does not adjust for the selection. The same
590 difference is -5.96 ($p = 0.23$) at ± 20 days and $+0.43$ ($p = 0.88$) at ± 60 . That window holds
591 113 postdating and 114 eligible cells, and the difference in mean standing is -2.52 with a
592 randomization p of 0.447. Cell counts in the 30 days either side of the cutoff stand at 80
593 against 84, a ratio of 0.95, so the running variable is not bunching against the threshold
594 [McCrary, 2008].

595 The mechanism is discontinuous where standing is not. The share of a cell’s window released
596 after the focal model jumps $+0.083$ at the cutoff, which at the eligible-cell gradient of -30.63
597 implies about -2.5 percentile points, inside every interval above, so at the cutoff the design
598 separates concealment from the availability gap only above that effect, and the narrowest
599 windows cannot reject the full -12.23 either. Peer count and raw score are continuous.
600 The global gap therefore lives in the 793 days that separate the two groups’ mean maturity
601 rather than in anything happening at the boundary itself.

602 A standing measure with no comparison window. The conclusion points at a scale esti-
603 mated jointly across benchmarks rather than against contemporaries, which removes the
604 comparison window from the measure entirely. Fitting one does not remove the gap. Abil-
605 ity and difficulty are estimated together over the 63 benchmarks reported on a common
606 zero-to-one scale, on a logit link, with the loading ridged toward one because a free load-
607 ing is unidentified at this density and takes slopes past 65,000. Restricting to benchmarks
608 carrying at least twenty releases and releases carrying at least three benchmarks leaves

1,669 eligible cells over 31 benchmarks. The model is fit on eligible cells alone, so the 226 postdating cells it can score are out of sample.

The eligible-minus-placebo gap in the residuals is +0.336 residual standard deviations, against +0.449 for the windowed percentile on the same contrast, positive in 64.0% of the 89 releases holding both kinds, at $p = 0.005$. Removing the window shrinks the gap by about a quarter and leaves it positive, which puts a jointly estimated scale beside trimming at 19% and side-balancing at 25%, each on its own scale, rather than ahead of them. Its difficulty parameters are estimated from the maturity-selected cells that exist, so part of the surviving gap may be the same selection re-expressed, and its placebo cells are scored out of sample while every eligible cell is in sample, an asymmetry the contrast inherits. Shuffling scores within benchmark, which leaves every date and the rank deficiency untouched, drops the same quantity to 0.02 on average over 30 draws.

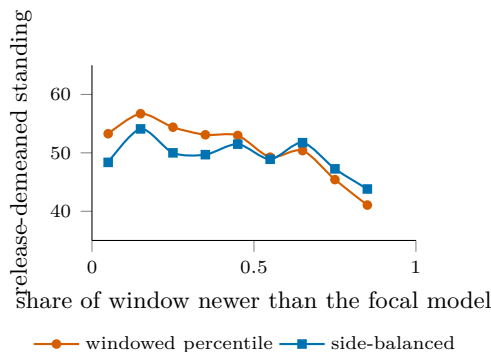


Figure 4: The side-balanced percentile against the windowed one, on eligible cells within release. The windowed measure slopes across the range while the side-balanced one is close to flat where cells are dense; the quoted gradients, -30.63 to -8.56 per unit share, are the within-release regression slopes on these cells, and the release-demeaned binned means shown are correspondingly flatter.

A backward-only standing, position among older peers alone, removes the window's forward half entirely and leaves the label-free contrast at +12.11 across 142 releases, so the gap does not come from look-ahead.

Drop-design ceiling and calibration. The panel contains 144 adjacent same-family release pairs covering 711 shared eligible benchmarks and 15 organisations. Requiring both releases in a pair to carry at least eight scored benchmarks leaves 37 pairs, 482 shared benchmarks, and 8 organisations, the effective cluster count, in the regime where Section 2.5 reports no analytic standard error. Under the null of no selective withholding, the sampling standard deviation of a drop gap, over 400 draws for each of 117 qualifying releases, is 19.4 percentile points when one benchmark is dropped, 14.4 when two are and 12.4 when three are. At conventional two-sided size a single-drop event must therefore move the gap by roughly 38 points before it separates from noise, three times the contamination itself, and no single drop can be read alone.

Possession evidence. Selective withholding is a statement about results a provider held and did not print, and the disclosure label observes only the printing. Four cuts organise how far the record reaches toward possession. The direct cases first: no document explicitly states that a benchmark was evaluated while reporting neither a score nor an outcome, so the direct-withholding category is empty over the 295 cells where a run is documented; the six named-without-number cells all carry an outcome in a figure, a chart, or a comparative claim. Second, the omission deficit stratified by evidence tier. Withholding predicts the deficit weakens, and eventually inverts, as possession evidence strengthens; composition predicts no gradient. The point estimates are +18.9 over 21 releases for dropped benchmarks, +11.3 over 51 for peer-reported omissions, and +11.3 over 69 for silent ones. The deficit does not become more negative as possession evidence strengthens; the dropped tier instead carries the largest positive deficit, a gradient that does not match the signature withholding predicts. Third, a change-based drop test. The estimator above compares levels; a sharper question is whether a benchmark is dropped after the family's standing on it deteriorates, which differences out persistent salience and relevance. Over the 42 family transitions with a scored

predecessor report, 33 dropped and 86 retained cells give a dropped-minus-retained change of -3.7 percentile points, the sign withholding predicts, at a count-preserving permutation $p = 0.16$ within transitions; on raw scores, kept where both sit on the unit interval, the gap is -0.7 points at $p = 0.71$. Both samples cut the 63 derived drop cells: the level estimator keeps the 16 sitting in releases that also retain at least two reported benchmarks to compare against, while the change-based test keeps the 33 with an independent score under both predecessor and successor and imposes no retention floor. Fourth, substitutions. Requiring the reporting table to hold its size while one benchmark is dropped and another inserted, the case a fixed-table explanation cannot cover, leaves no qualifying case in the record. Nothing in the four detects withholding. The change-based gap is directionally consistent with it and statistically indistinguishable from its permutation null; every other cut is null, absent, or opposite in sign.

Reliability detail. The second coding covers 20 of the 100 readable releases, drawn stratified by provider. Derived categories import contemporaries' evidence across releases, a dependence the release resampling does not model. Across 15 of them, 194 derived cells pair: one drawn release is a zero-disclosure document whose blank evidence rows the pairing drops, a conservative exclusion since all eleven of its cells are trivial agreements, and four drawn releases sit outside the pinned panel. The reported-versus-not cross-tabulation is one-directional, 47 jointly reported and 143 jointly not: no cell the first coder called reported was coded otherwise by the second, and the four reversals all sit on one release whose second-coding note records retaining four contested scores despite the first coder's disagreement, an exposure that breaks blindness for that release. Excluding it, reported-versus-not agreement is exact and the full-category κ is 0.88 on 182 cells. Of the 14 disagreements overall, seven are the variant boundary between C and A, one a bare mention against a value, two the derived split between G and H on a single release, and four the exposed release above. Three limits of the record are candidly unrepaired: the eight releases without a locatable document are unlikely to be missing at random and plausibly document less; the evaluator's coverage may itself respond to what providers emphasise, which would tilt the reported set upward by sampling; and the best-across-scaffolds score rule is not probed for asymmetry across the boundary. High-suspicion marginals match exactly at 5.7% for each coder, which is why the split κ holds at 0.81 despite the category's rarity; its wide interval reflects eleven such cells concentrating in a few releases.

Provider-only clustering, for comparison. The manuscript's regressions cluster on provider and benchmark jointly. Provider-only errors are consistently smaller: 1.271 against 2.038 on the release-effects row, a ratio of 1.60, and 1.652 against 1.748 on the fully saturated row, a ratio of 1.06. The benchmark dimension carries real dependence on the raw contrast and almost none once benchmark effects and the window terms are in, which is what the decomposition says it should. The headline sign-flip test at $p = 0.0001$ preserves provider dependence only; the two-way regression row above is the statement that the contamination survives both dimensions, at $t = -6.6$.

689 NeurIPS Paper Checklist

690 1. Claims

691 Question: Do the main claims made in the abstract and introduction accurately
692 reflect the paper’s contributions and scope?

693 Answer: [\[Yes\]](#)

694 Justification: The abstract and introduction state both findings within their identi-
695 fied scope: the coded omission gap is indistinguishable from the label-free gap, and
696 within-release composition tracks standing. Section 5 reports the reliability of the
697 coding behind them.

698 2. Limitations

699 Question: Does the paper discuss the limitations of the work performed by the
700 authors?

701 Answer: [\[Yes\]](#)

702 Justification: Limitations appear where they bear on the argument rather than
703 only at the end. Section 2.1 bounds the estimand and states that the reported
704 omission rate is descriptive and identifies no withholding; Section 3.2 states that
705 the conditioning is a decomposition rather than a control strategy, and what a
706 study holding disclosure labels would additionally have to argue; Section 3.3 reports
707 the corrections that fail and the coverage the balanced measure loses; Section 5
708 reports the ceiling on the design that narrows endowment uncertainty furthest, its
709 16 realised drops, and the underpowered null they return.

710 3. Theory assumptions and proofs

711 Question: For each theoretical result, does the paper provide the full set of assump-
712 tions and a complete (and correct) proof?

713 Answer: [\[Yes\]](#)

714 Justification: Section 2.1 states one identification result, that a temporal eligibility
715 gate cannot satisfy the exclusion restriction the design needs, and gives its argument
716 in full: the three date variables are exactly collinear, so excluding one requires
717 that at least two of them not move standing. Section 3 shows all three move
718 it. Appendix B states the three sign restrictions used to bound the unidentified
719 component and derives the bound from them. Everything else in Section 2.1 restates
720 published results with citation.

721 4. Experimental result reproducibility

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723 the main experimental results of the paper to the extent that it affects the main
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730 key, the eligibility rule, the window width, and which estimates impose a minimum
731 benchmark count and which do not.

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734 instructions to faithfully reproduce the main experimental results, as described in
735 supplemental material?

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738 dent scores are redistributed by their publisher under CC-BY and are fetched rather
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6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: There is no model training. The choices that affect the estimates are stated in Section 2: the release key, the rule for collapsing scaffold variants, the 182-day comparison window, which estimates impose a minimum benchmark count and which do not, and the treatment of pairs whose benchmark vintage is unverified.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 2.5 states the inference protocol. The cluster count is small and unevenly distributed, so below twelve clusters no analytic standard error is reported, coefficients in that range use a restricted wild cluster bootstrap, release-level means use a cluster sign-flip randomization test, and no result is called significant on a cluster-t alone. Table 1 reports a standard error with every coefficient, the headline contrast carries a randomization p-value, and the permutation null is reported with its mean, its spread and the number of draws reaching the observed value.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

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Justification: The work uses published benchmark scores and public release documents, involves no human subjects and no personal data, and makes no claim about any individual.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 4 states the consequence for a reader of a pool-relative leaderboard, namely that such a number describes standing within one vintage of the evaluated pool rather than a property of the model. The conclusion states that whether the record needed to measure withholding should grow is a reporting-standard question. The paper does not attribute concealment to any named provider, because the design does not support that.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

792 Answer: [N/A]
793 Justification: No model, dataset or other asset with misuse potential is released.
794 The release is analysis code and derived tables built from already public sources.

795 12. Licenses for existing assets
796 Question: Are the creators or original owners of assets (e.g., code, data, models),
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802 named in the text and in the released code.

803 13. New assets
804 Question: Are new assets introduced in the paper well documented and is the
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806 Answer: [Yes]
807 Justification: The released code documents how each number is produced, the snap-
808 shot manifest records the exact data build, and the frozen external evidence records
809 a source URL, a retrieval date and a SHA-256 for every file.

810 14. Crowdsourcing and research with human subjects
811 Question: For crowdsourcing experiments and research with human subjects, does
812 the paper include the full text of instructions given to participants and screenshots,
813 if applicable, as well as details about compensation (if any)?
814 Answer: [N/A]
815 Justification: The work involves no crowdsourcing and no human subjects.

816 15. Institutional review board (IRB) approvals or equivalent for research with human
817 subjects
818 Question: Does the paper describe potential risks incurred by study participants,
819 whether such risks were disclosed to the subjects, and whether Institutional Review
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821 of your country or institution) were obtained?
822 Answer: [N/A]
823 Justification: The work involves no human subjects, so no review board approval
824 was required.

825 16. Declaration of LLM usage
826 Question: Does the paper describe the usage of LLMs if it is an important, original,
827 or non-standard component of the core methods in this research? Note that if the
828 LLM is used only for writing, editing, or formatting purposes and does not impact
829 the core methodology, scientific rigor, or originality of the research, declaration is
830 not required.
831 Answer: [Yes]
832 Justification: All disclosure-coding categories standing in the paper were hand-
833 assigned by a human coder from archived release-time documents under the frozen
834 protocol, with an independent second coder recoding a provider-stratified 20% sam-
835 ple (Section 5), so no number reported here depends on a model’s judgment. A
836 superseded early pass had used a large language model on the same task; those
837 labels were cleared before analysis and none enters. Every number is produced by
838 the released code from the pinned snapshot. Language models were otherwise used
839 only for writing and code assistance.